Chemistry GCE O Level 2021 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0515 Chemistry 1, GCE Ordinary Level, June 2021, one and a half hours). Answer ALL 50 questions; all questions carry equal marks. Each question has FOUR suggested answers A, B, C and D; choose the one that is appropriate. Non-programmable calculators are allowed. Questions 29-31 and 45-46 use special response formats explained with the questions.

Relative atomic masses: N = 14.0, H = 1.0, O = 16.0, C = 12.0, Na = 23.0, Mg = 24.0; Avogadro number = 6.0×10^{23}.

Multiple choice questions - choose the one correct answer, A, B, C or D.

1. Name a metal that is extracted by thermal reduction

a) Aluminium
b) Copper
c) Iron
d) Sodium

2. Oxidation can be defined as

a) The gain of electrons
b) Addition of hydrogen
c) Loss of oxygen
d) Addition of oxygen

3. Identify the change of state from solid to gas

a) Sublimation
b) Evaporation
c) Melting
d) Fusion

4. Which of the following is the main component of inactive air?

a) Argon
b) Nitrogen
c) Oxygen
d) Carbon dioxide

5. Identify a species which is an isotope of $^{35}_{17}X$

a) $^{35}_{17}X$
b) $^{35}_{18}X$

c) $^{38}_{17}X$
d) $^{36}_{18}X$

6. Explain why hydrogen gas is collected by upward delivery

a) It is heavier than air
b) It easily displaces air
c) It is lighter than air
d) Insoluble in water

7. Which of the following anions will form a precipitate with barium chloride in acid medium?

a) Sulphate ion
b) Carbonate ion
c) Sulphite ion
d) Nitrate ion

8. How many molecules are there in 4.5 g of water? (RMM of water = 18)

a) 0.25×10^{23} molecules
b) 25.5×10^{23} molecules
c) 4.5×10^{23} molecules
d) 1.5×10^{23} molecules

9. Identify the gas that naturally makes rain slightly acidic

a) Sulphur dioxide
b) Carbon dioxide
c) Hydrogen sulphide
d) Nitrogen dioxide

10. Select a dibasic acid from the list below

a) HCl
b) $\mathrm{HNO_3}$
c) $\mathrm{H_2SO_4}$
d) $\mathrm{CH_3COOH}$

11. Determine the number of neutrons in an atom of $^{40}_{18}A$

a) 18 neutrons
b) 40 neutrons
c) 58 neutrons
d) 22 neutrons

12. For the reaction

$$\mathrm{CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}}$$

The rate of reaction can be increased by

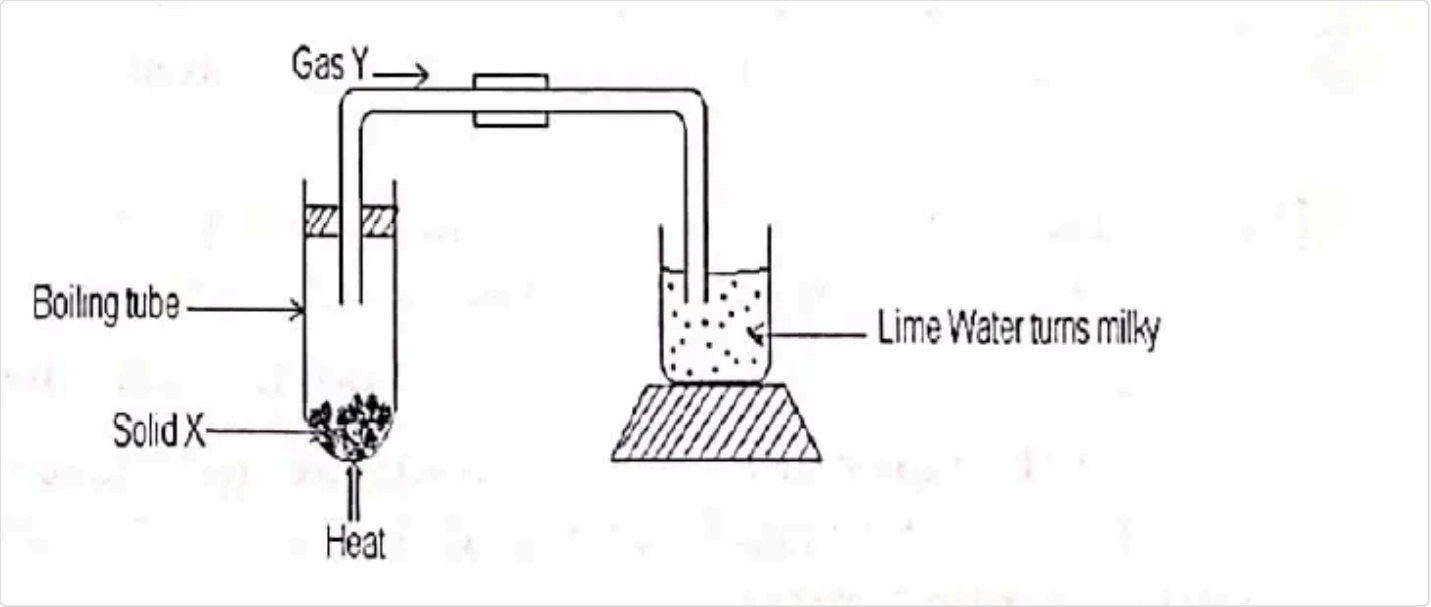
a) Placing the reaction mixture in an ice bath
b) Grinding the $\mathrm{CaCO_3}$

c) Reducing the concentration of HCl
d) Increasing the pressure

13. Which of the following salts will give a brick red flame?

a) $CaCO_3$
b) Na_2CO_3
c) $BaSO_4$
d) K_2SO_4

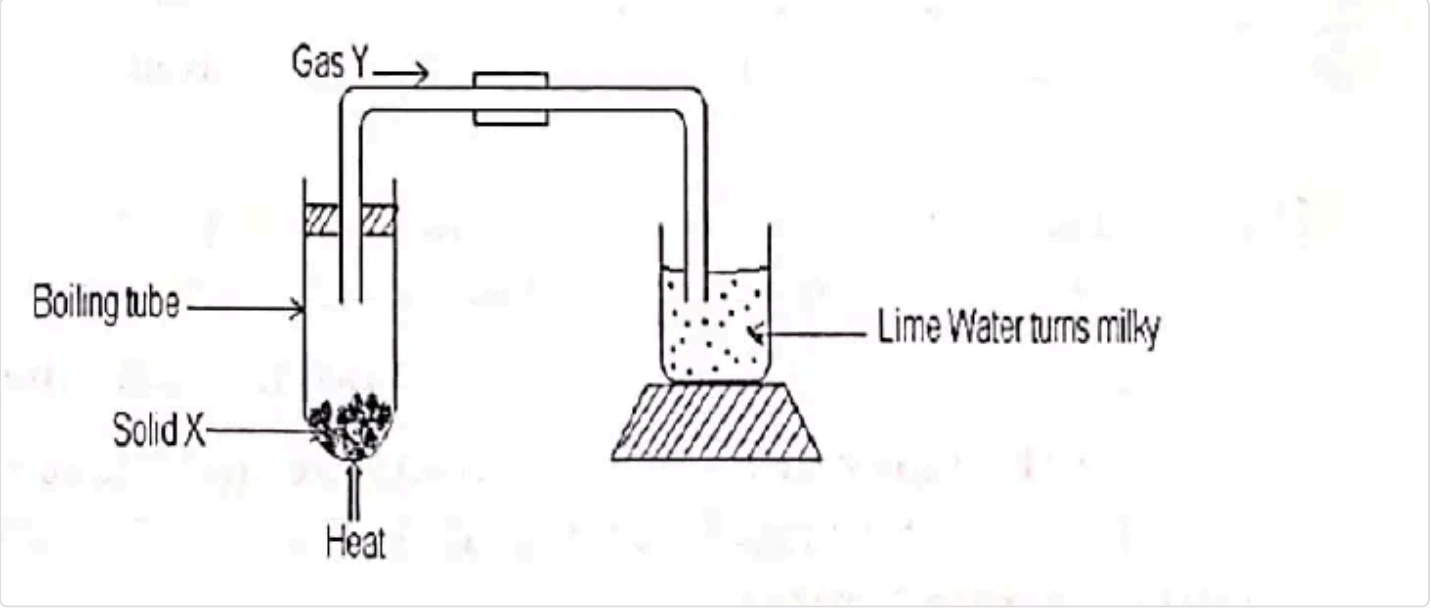
14. Questions 14 and 15 concern the following diagram



Identify the solid S

a) Sodium carbonate
b) Sodium sulphite
c) Calcium oxide
d) Calcium carbonate

15. Questions 14 and 15 concern the following diagram



Why does the lime water turn milky?

a) Solid S is white in colour
b) An insoluble salt is formed
c) A white solution is formed
d) A soluble salt precipitates

16. Compound X is bubbled through a solution of aqueous bromine, the reddish brown solution is decolourised. X could be

a) $\mathrm{C_2H_6}$
b) $\mathrm{CH_3COOH}$
c) $\mathrm{C_2H_5OH}$
d) $\mathrm{C_3H_6}$

17. Which of these contains the same number of particles?

a) 1 mole of water and 40 g of NaOH
b) 1.2 g of Carbon and 18 g of water
c) 2.4 g of magnesium and 17 g of ammonia
d) $2400\ \mathrm{cm^3}$ of $\mathrm{O_2}$ and 2 g of hydrogen

18. What is the colour change when 2 drops of phenolphthalein are added to a solution of sodium hydroxide?

a) Pink to colourless
b) Pink
c) Colourless to pink
d) Red to blue

19. Using the equation

$$\mathrm{C_2H_{4(g)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(l)}}$$

Calculate the heat evolved when 2.8 g of $\mathrm{C_2H_4}$ are completely burnt. $\Delta H = -1410\ \mathrm{kJ\,mol^{-1}}$

a) 2820 kJ
b) 3948 kJ
c) 503.6 kJ
d) 141 kJ

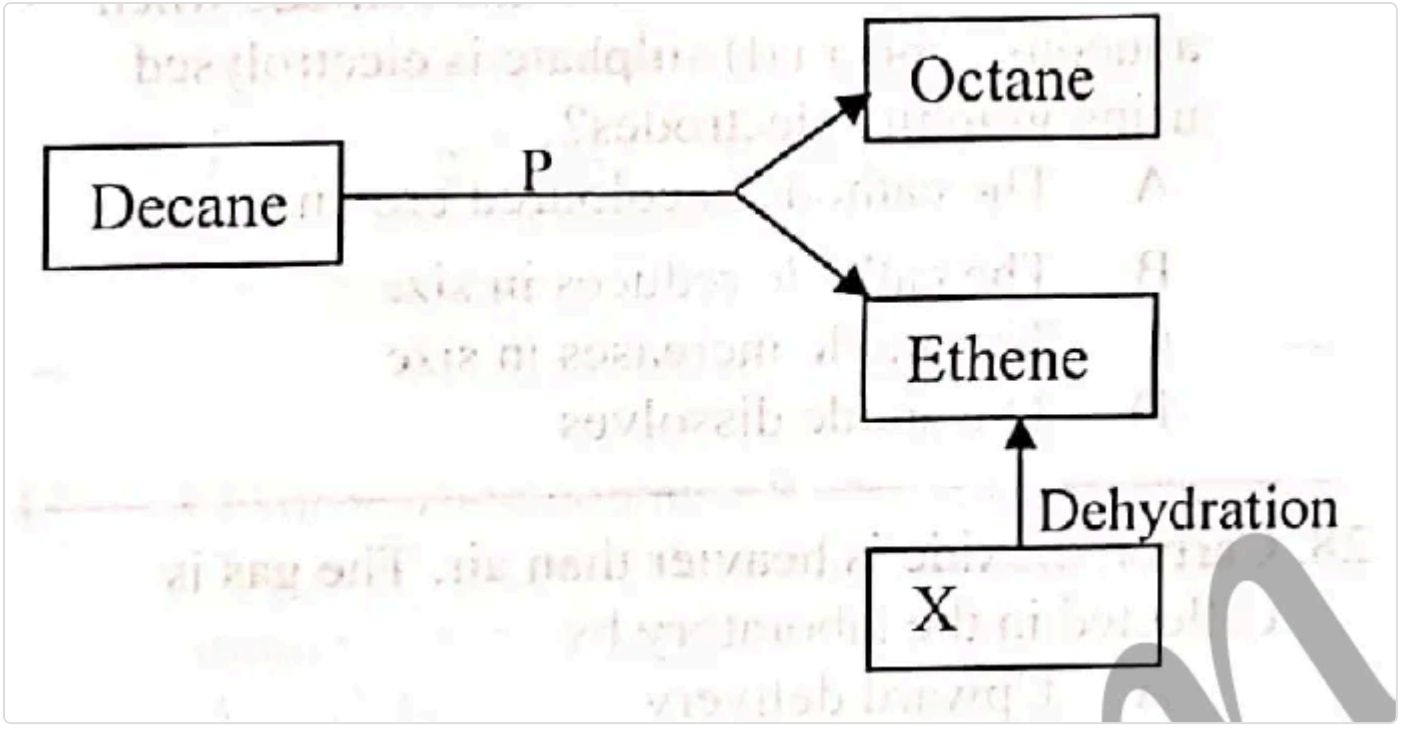
20. Name the apparatus that is used in the laboratory to measure the temperature of a liquid

a) Measuring cylinder
b) Thermometer
c) An Ammeter
d) A Burner

21. The compound ammonium sulphate is commonly used as

a) A fungicide
b) A cleansing agent
c) A fertilizer
d) An antiseptic

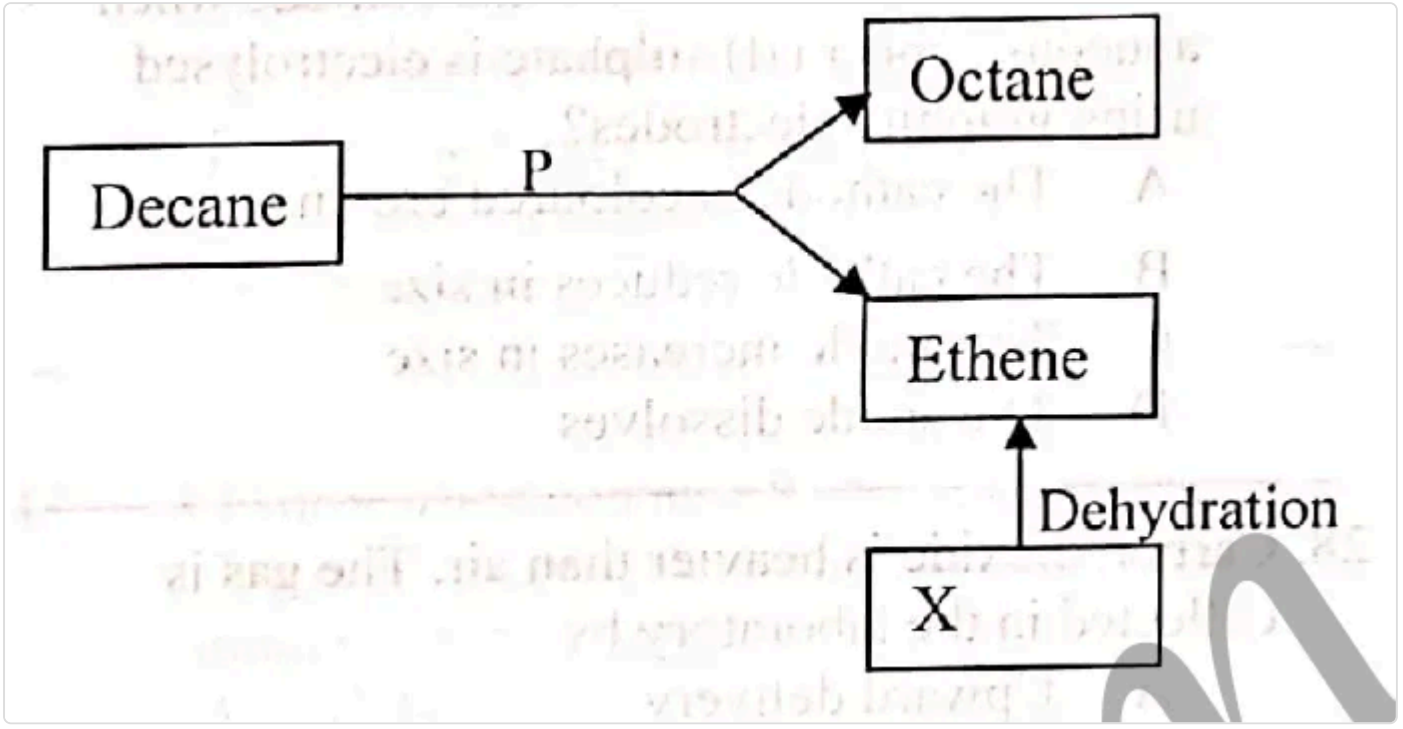
22. Questions 22-23: concern the following diagram.



Identify the process, P.

a) Cracking
b) Distillation
c) Decomposition
d) Addition

23. Questions 22-23: concern the following diagram.



Identify the compound X

a) Ethane
b) Ethyne

c) Ethanoic acid
d) Ethanol

24. The following reaction is used to manufacture NH_3

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ Δ -ve

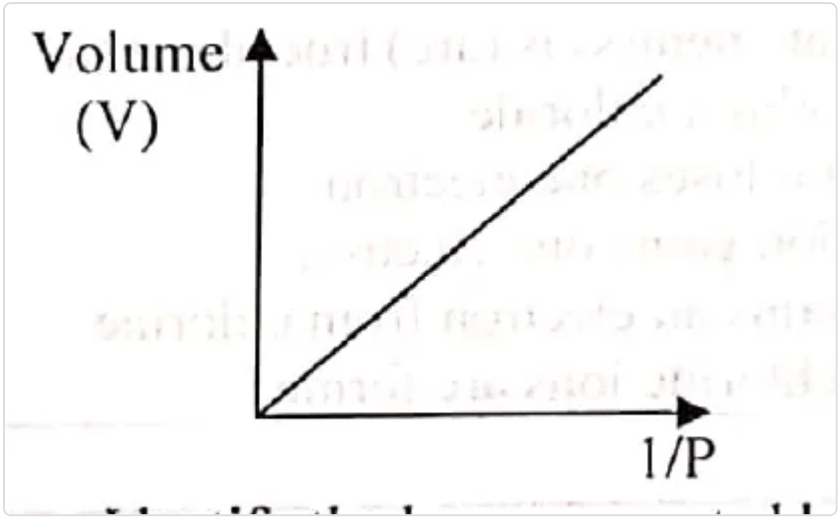
Which of these changes would increase the yield of NH_3?

a) Decreasing the pressure
b) Increasing temperature
c) Using a catalyst
d) Increasing pressure

25. Identify a metal which is used to make kitchen utensils and also in the construction of Aircrafts

a) Cu
b) Al
c) Ti
d) Mg

26. The behaviour of some gases is represented by the following graph



Identify the law represented by the graph

a) Gas law
b) Charles' law
c) Boyle's law
d) Avogadro's law

27. What would you observe at the cathode when aqueous copper (II) sulphate is electrolysed using graphite electrodes?

a) The cathode is coloured brown
b) The cathode reduces in size
c) The anode increases in size
d) The anode dissolves

28. Carbon dioxide is heavier than air. The gas is collected in the laboratory by

a) Upward delivery
b) Downward delivery
c) Downward displacement
d) Passing over water

29. For each of the questions, ONE or MORE of the response(s) given is (are) correct. Decide which of the response(s) is (are) correct. Then choose:
A. If 1, 2 and 3 are correct; B. If 1 and 3 are correct; C. If 2 and 4 are correct; D. If 4 only is correct.

Which of the following equation (s) represent(s) thermal Decomposition

1. $\mathrm{NH_4Cl_{(s)} \rightarrow NH_{3(g)} + HCl_{(g)}}$
2. $\mathrm{Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + Cl_{2(g)} + H_2O_{(l)}}$
3. $\mathrm{CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}}$
4. $\mathrm{NaCl_{(aq)} + H_2O_{(l)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)}}$

a) 1, 2 and 3 only
b) 1 and 3 only
c) 2 and 4 only
d) 4 only

30. For each of the questions, ONE or MORE of the response(s) given is (are) correct. Decide which of the response(s) is (are) correct. Then choose:
A. If 1, 2 and 3 are correct; B. If 1 and 3 are correct; C. If 2 and 4 are correct; D. If 4 only is correct.

Select which statement(s) is (are) true about the formation of sodium chloride

1. the sodium ion loses one electron
2. the chloride ion gains one electron
3. the sodium gains an electron from chlorine
4. sodium and chloride ions are formed

Options as printed: A) 1, 2 and 3 only B) 1 and 3 only C) 2 and 4 only D) 4 only

31. For each of the questions, ONE or MORE of the response(s) given is (are) correct. Decide which of the response(s) is (are) correct. Then choose:
A. If 1, 2 and 3 are correct; B. If 1 and 3 are correct; C. If 2 and 4 are correct; D. If 4 only is correct.

Which of the following industrial processes involves the use of electrolysis?

1. Electroplating
2. Manufacture of NaOH
3. Extraction of Al
4. Extraction of Cu

a) 1, 2 and 3 only
b) 1 and 3 only
c) 2 and 4 only
d) 4 only

32. State one common property of metals

a) They are brittle
b) They are soft
c) They conduct electricity
d) They have more than one valency

33. A compound, X, gives a golden yellow flame. On addition of dilute hydrochloric acid to the compound, a gas is evolved that turns lime water milky. Identify the compound X

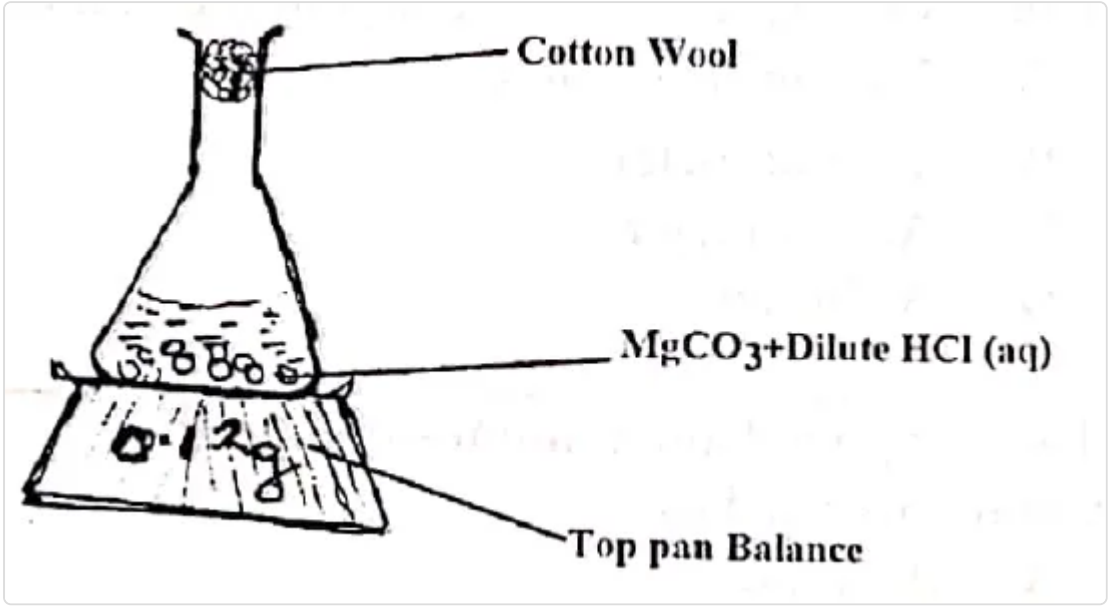
a) K_2SO_3
b) Na_2CO_3
c) Na_2SO_4
d) K_2CO_3

34. Determine the molar mass of ammonium Nitrate (NH_4NO_3)

a) 66 g/mol
b) 122 g/mol
c) 80 g/mol
d) 45 g/mol

35. Questions 35-36

The following setup was used to study the reaction between magnesium carbonate and dilute hydrochloric acid

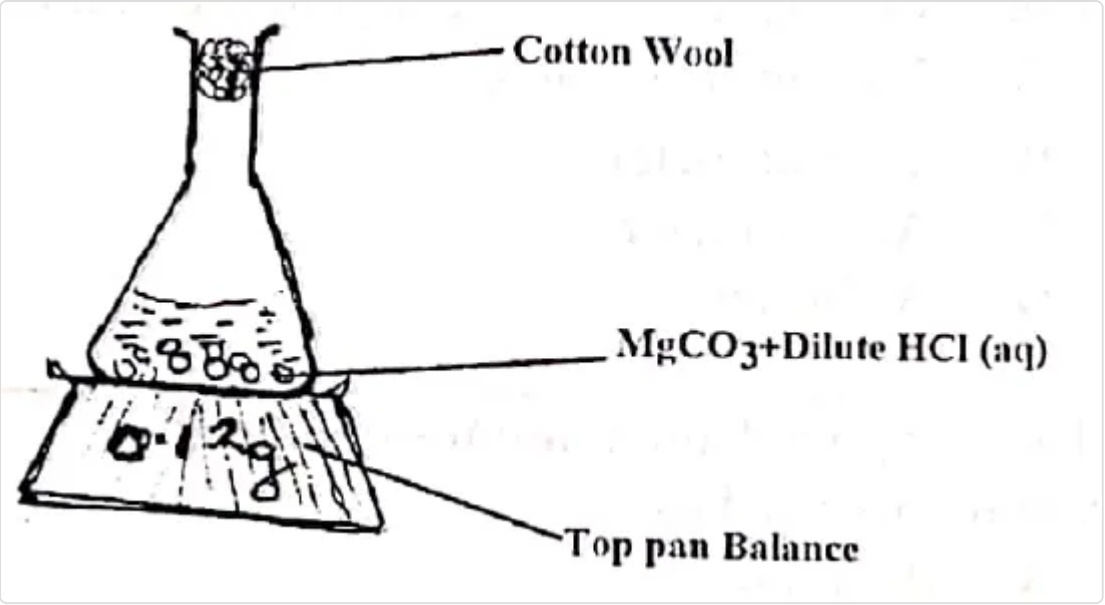


Why is there a gradual loss in mass?

a) The carbonate dissolves
b) The carbonate decomposes
c) Less product is formed
d) CO_2 escapes

36. Questions 35-36

The following setup was used to study the reaction between magnesium carbonate and dilute hydrochloric acid



Why is dilute sulphuric acid not suitable for use in this experiment?

Options as printed: A) Sulphuric acid forms an insoluble salt B) Sulphuric acid is less reactive C) Sulphuric acid is very soluble D) Sulphuric acid will oxidise the carbonate

37. Which of the following substances will form a coloured precipitate with aqueous sodium hydroxide?

a) Na_2SO_4
b) $CaSO_4$
c) $PbSO_4$
d) $CuSO_4$

38. Why is sulphur trioxide not dissolved directly in water in the manufacture of sulphuric acid?

a) Sulphur trioxide is not very soluble in water
b) The reaction is violent and highly exothermic
c) Some of the gas will escape undissolved
d) It will take a longer time for the acid to be produced

39. In an acid/ base titration, $20\ cm^3$ of dilute hydrochloric acid were required to neutralise $25\ cm^3$ of 0.1 M sodium carbonate solution. Determine the molarity of the dilute hydrochloric acid.

a) 0.01 M
b) 0.13 M
c) 0.25 M
d) 2 M

40. Questions 40 - 41

Two elements W and V have electronic configuration W = 2: 8: 1 V = 2: 6

Write the formula of the compound formed between W and V

a) W_2V
b) WV
c) WV_2
d) WV_6

41. Questions 40 - 41

Two elements W and V have electronic configuration W = 2: 8: 1 V = 2: 6

Write an equation for the reaction between hydrogen and V

a) $2H_2 + V_2 \rightarrow 2H_2V$
b) $2H + V \rightarrow H_2V$
c) $H_2 + V_6 \rightarrow 2HV_3$
d) $H_2 + V_6 \rightarrow H_2V_6$

42. Which of the following is a mixture?

a) Common salt
b) Sodium bicarbonate
c) Liquid ammonia
d) Sea water

43. A hydrocarbon contains 9.25 g of carbon and 0.75 g of hydrogen. Determine the empirical formula of the compound

a) CH_2
b) CH
c) C_2H_2
d) C_2H_4

44. Which of the following is the best procedure for diluting an acid in the laboratory?

a) The water is poured into a beaker containing acid
b) The water is added to a known volume of acid
c) A known volume of acid is added to water
d) The water is warmed then added to the acid

45. Questions 45-46

Each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether each of the statements is TRUE OR FALSE. Then mark:
A: if both statements are true and the second statement is a correct explanation of the first statement; B: if both statements are true but the second statement is not a correct explanation of the first statement; C: if the first statement is true but the second is false; D: if the first statement is false but the second statement is true.

	Statement 1	Statement 2
45	1 mole of carbon contains 6.02×10^{23} atoms	The amount of substance in 1.2 g of carbon is 0.1 mol

a) Both true, and the second statement is a correct explanation of the first
b) Both true, but the second statement is NOT a correct explanation of the first
c) First statement true, second statement false
d) First statement false, second statement true

46. Questions 45-46

Each of the following questions consists of a statement in the left-hand column followed by a second statement in the right-hand column. Decide whether each of the statements is TRUE OR FALSE. Then mark:
A: if both statements are true and the second statement is a correct explanation of the first statement; B: if both statements are true but the second statement is not a correct explanation of the first statement; C: if the first statement is true but the second is false; D: if the first statement is false but the second statement is true.

	Statement 1	Statement 2
46	Hydrogen gas is prepared by action of copper metal on dil HCl	Copper is below hydrogen on the electrochemical series

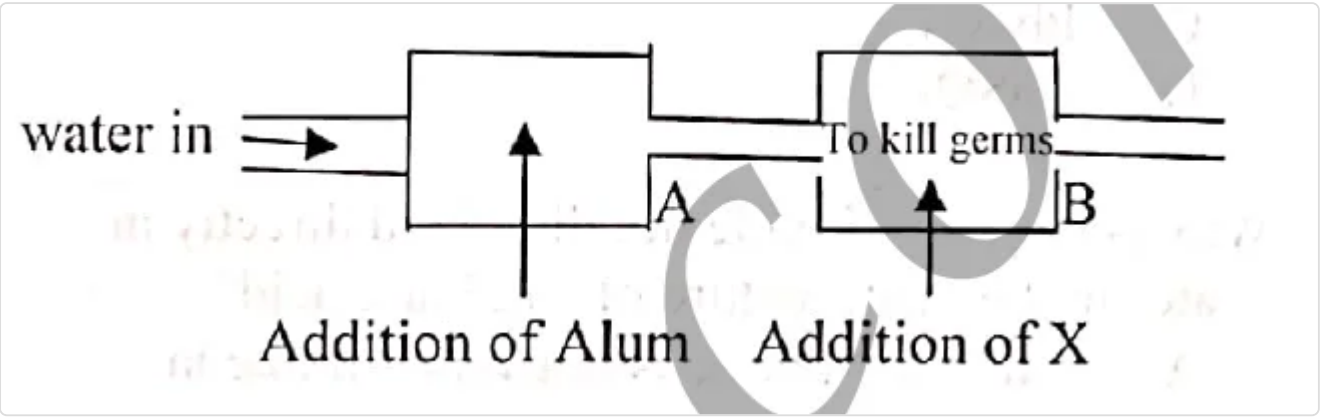
a) Both true, and the second statement is a correct explanation of the first
b) Both true, but the second statement is NOT a correct explanation of the first
c) First statement true, second statement false
d) First statement false, second statement true

47. Which of the following organic compounds would evolve white fumes with PCl_5?

a) C_2H_5OH
b) CH_3OCH_3
c) CH_3COOCH_3
d) C_2H_4

48. Questions 48-49

The following diagram shows some of the steps used in the purification of water for domestic use.

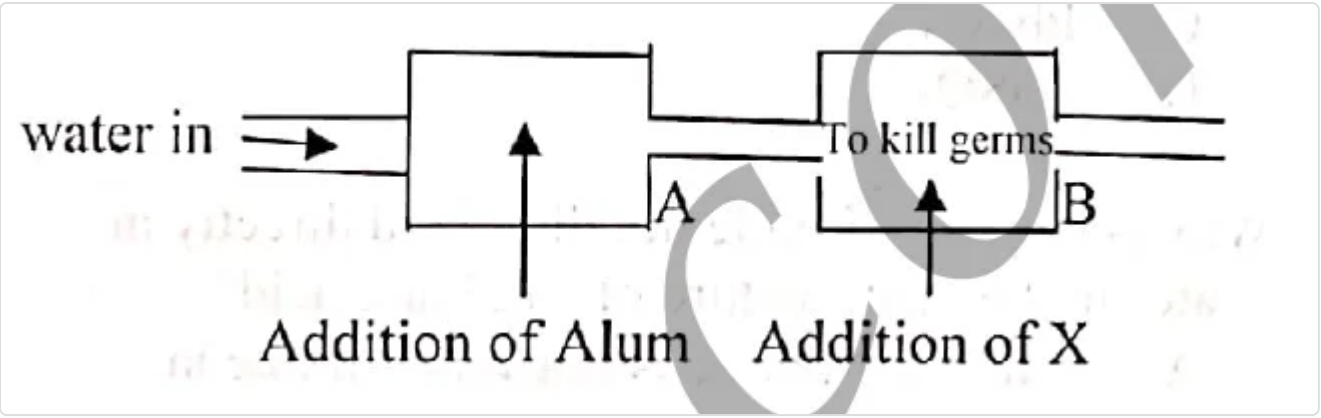


Why is alum added in tank A?

a) To remove soluble impurities in water
b) To coagulate tiny particles in suspension
c) To remove smell and taste from water
d) To increase the pressure of water into tank B

49. Questions 48-49

The following diagram shows some of the steps used in the purification of water for domestic use.



What is the substance used to kill germs in tank B?

a) Fluorine
b) Calcium hydroxide
c) Chlorine
d) Ammonium hydroxide

50. Identify the substance in air that supports Combustion

a) Oxygen
b) Nitrogen
c) Argon
d) Carbon dioxide

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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